CLASS 9 · CBSE · GANITA MANJARI PART 2 · DPP SHEET

Chapter 10: How Quantities Combine: Understanding Data

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Section A has 25 students with average marks 64, and Section B has 15 students with average marks 72. Find the combined average.
2. Three groups of the same size have averages 10, 14 and 18. Find the combined average.
3. 200 mL of a 5% salt solution is mixed with 300 mL of a 10% salt solution. Find the concentration of the mixture.
4. Scores of 50, 60 and 80 are combined in the ratio $1 : 2 : 2$. Find the weighted mean.
5. The average of 9 numbers is 20. A tenth number, 40, is added. Find the new average.
6. The average of 12 numbers is 15. The number 26 is removed. Find the average of the rest.
7. How much water must be added to 4 litres of a 15% sugar solution to make it a 10% solution?
8. A café is rated 4 for food, 2 for service and 5 for ambience, with weights $3 : 2 : 1$. Find the overall rating.
9. A family spends ₹6000 on food, ₹4000 on rent and ₹2000 on other things. Find the percentages for a 100% stacked bar.
10. True or false: a 100% stacked bar chart shows which bar has the larger total.

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Answer Key

1. $\frac{1600 + 1080}{40} = 67$
2. 14
3. $\frac{10 + 30}{500} = 8\%$
4. $\frac{50 + 120 + 160}{5} = 66$
5. $\frac{220}{10} = 22$
6. $\frac{154}{11} = 14$
7. 2 litres
8. $\frac{12 + 4 + 5}{6} = 3.5$
9. 50%, 33.3% and 16.7%
10. False. Every bar is drawn to the same length, so the totals cannot be read from it.

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